

Breast Cancer Detection with a Neural Network

Portfolio Project Report

Objective: classify tumors as malignant (M) or benign (B) using the supplied Wisconsin Diagnostic Breast Cancer CSV, with evaluation centered on malignant-case sensitivity and false-negative control.

Dataset & preprocessing

The supplied CSV contains 569 records. The project uses 30 numeric tumor measurements. The identifier column and the empty Unnamed: 32 column are removed. M is encoded as the positive class (1) and B as 0. A stratified train/validation/test split is used, and standardization is fitted only on training data to prevent leakage.

Modeling approach

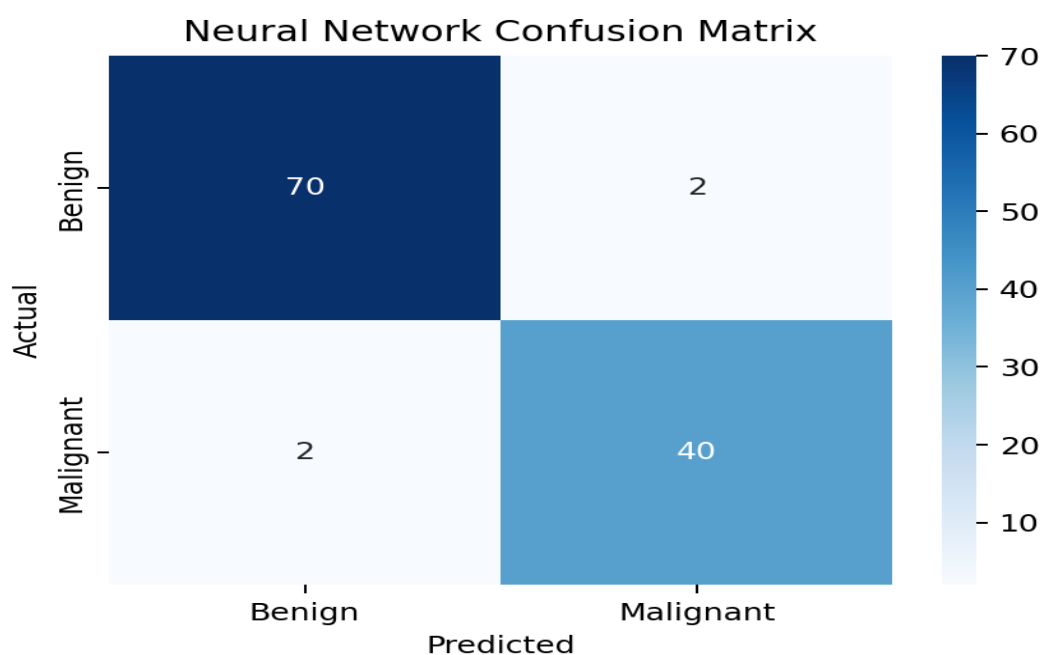
The primary model is a PyTorch neural network using dense layers, BatchNorm, ReLU activations, dropout and AdamW optimization with early stopping. Logistic Regression, Random Forest and XGBoost are used as transparent performance benchmarks.

Held-out test results

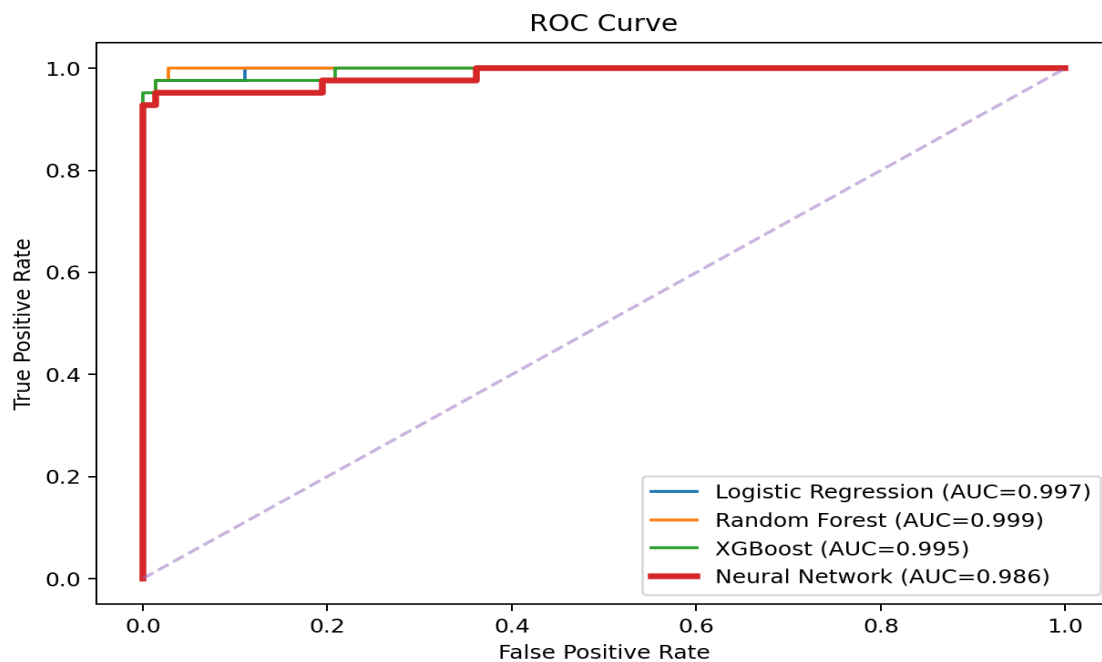
Model	Accuracy	Precision	Sensitivity	Specificity	F1	ROC-AUC	PR-AUC
Logistic Regression	0.982	0.976	0.976	0.986	0.976	0.997	0.995
Random Forest	0.974	1.0	0.929	1.0	0.963	0.999	0.998
XGBoost	0.974	1.0	0.929	1.0	0.963	0.995	0.993
Neural Network	0.965	0.952	0.952	0.972	0.952	0.986	0.984

Neural Network headline: 96.49% accuracy, 95.24% sensitivity, 97.22% specificity and 0.986 ROC-AUC on the held-out test set.

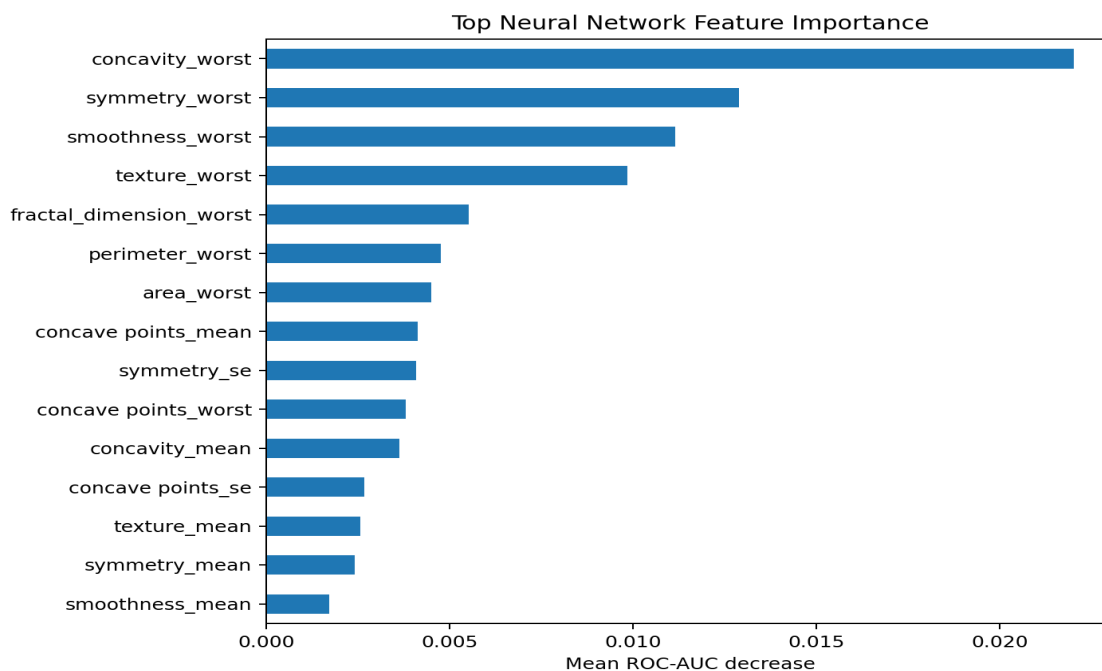
Confusion Matrix



ROC Curve



Top Feature Importance



Responsible-use note

This project is an educational demonstration, not a clinical diagnostic system. Real-world deployment would require external validation, calibration assessment, prospective studies, clinical oversight, regulatory review, robust data governance and monitoring for bias and distribution shift.